Two reaction–diffusion species driving growth and death:

is the mechanism sound as biology, as code, and as Plexus2?

`tyssue_rd_ops` · `tyssue_ops3d` · `make_two_species.py` · `make_sculpt.py` · `log/okuda/{ts,tsd,sc}_*` (24 preliminary runs) · 11 August 2026

Scope. This note is about one addition to the framework: a second reaction–diffusion species, and the two operators that consume it — growth inhibition and cell death. It asks whether that addition is *sound*, on three axes: as biology, as code, and as Plexus2. **It does not ask whether the mechanism achieves the campaign's goal.** Twenty-four preliminary runs exist in `log/okuda` and they raised a specific set of defects; §6 is that list, structured, with each one's status. A mechanism whose implementation is unverified cannot be compared with an observation at all, so the goal question is deferred until this note's gates close.

1. The mechanism

One field cannot pattern two processes. With a single Gray–Scott pair every operator reads the same activator — growth is gated on it, death is gated on it, the purse-string selects on it — so the tissue carries exactly one map. "Where it grows" and "where it dies" are then the same region or exact complements of it, by construction, and no setting of any parameter separates them. This is a property of the representation.

Two species is two maps. Biologically this is unremarkable and well attested: a tissue carries many morphogens at once, on different length scales, read by different targets. Two are enough to state the smallest interesting case — one field says *grow here*, another says *stop* or *die here* — and that is the case this note verifies.

The two consumers are drawn from different literatures. Growth inhibition is lateral inhibition in the Gierer–Meinhardt sense: a diffusible signal that suppresses the response of the cells around a source. Patterned death is Monier *et al.* (2015), where apoptotic force drives epithelial folding in *Drosophila* — and it is the only *inward* deformation available in this vocabulary, every other operator pushing the sheet outward.

The arithmetic that fixes the representation. A dying cell leaves an epithelium by losing neighbours until it is a triangle; a triangle collapses to a point as a T2 transition, $V-2$, $E-3$, $F-1$, so $\chi = V-E+F$ is unchanged and the sheet stays closed at genus 0. Collapsing a face with $k>3$ sides would merge k vertices into one and leave a vertex of degree k, **which a trivalent sheet cannot represent**. So death cannot be one operator: it is mark, then shed, then extrude, and the shedding belongs to `reconnect_t1_3d`. That is not a hidden coupling; it is the mechanism. A cell leaves an epithelium by losing neighbours.

Two ratios fix the proportions. The species' wavelengths must differ or the second map is redundant: measured on a static sphere, species A at $\chi = 1.3$ holds 2 spots at a spacing of 6.99 cell diameters and species B at $\chi = 0.4$ holds 15 at 3.18 — a factor of 2.2, with **A** the coarse map. χ sets this, not the diffusivity pair: at $\chi = 1.3$ with doubled diffusivities the field does not pattern at all. And the clearing time of a dying cell is fixed by two numbers and nothing else (Eq. (4)).

2. Equations and symbols

Two independent systems in one buffer. `chem` is width 4; species A occupies columns $(0,1)$ and species B columns $(2,3)$. Each has its own seeder, diffusion and reaction, and each operator declares which pair it owns through `chan`:

$$\frac{\partial a_X}{\partial t} = D_a^X \nabla^2 a_X + f(a_X, u_X), \qquad \frac{\partial u_X}{\partial t} = D_u^X \nabla^2 u_X + g(a_X, u_X), \qquad X \in \{A, B\}, \tag{1}$$

with f, g Gray–Scott. The two systems are coupled *only* through the operators that read them, never through the buffer — which is a property that must be gated (G7), not assumed.

Growth, and the zero it did not have. The standing law is purely activating,

$$\dot{v}_{\text{eq}} = \texttt{rate} \cdot \big(\rho + \text{hill}(a_A)\big), \qquad \text{hill}(x) = \frac{x^n}{x_{\text{sw}}^n + x^n} \in [0,1], \tag{2}$$

so the slowest a cell can grow is $\texttt{rate} \cdot \rho$, and $\rho > 0$ is required by premise P1 (a tissue that adds no material is not growing). The inhibitor supplies the missing zero, multiplicatively:

$$\dot{v}_{\text{eq}} = \texttt{rate} \cdot \big(\rho + \text{hill}(a_A)\big) \cdot \big(1 - \text{hill}_{\text{inh}}(\hat{a}_B)\big), \qquad \hat{a}_B = a_B / \max_f a_B. \tag{3}$$

$\hat{a}_B$ is normalised to the inhibitor's own maximum for the same reason a_{sw} is a fraction: a threshold stated absolutely, against a field whose scale the chemistry sets, is either always on or always off.

Death, as three steps and one closed-form time.

$$V_f^0 \leftarrow (1-s)V_f^0, \qquad \text{extrude when } V_f^0 < c\,v_{\text{ref}}, \qquad \boxed{\tau_{\text{clear}} = \frac{\ln c}{\ln(1-s)} \text{ ticks}} \tag{4}$$

Volume is conserved into the survivors, and the conservation is bounded. When f dies its remaining material is shared over its live ring neighbours, and the increment is capped so the recipient stays inside the integrator's basin:

$$c_g \leftarrow c_g + \min\!\Big(\frac{\text{amt}/|\mathcal{L}|}{V_g^0},\ c_{\max} - c_g\Big), \qquad \mathcal{L} = \{g \in \mathcal{N}(f) : V_g^0 \geq c\,v_{\text{ref}}\}, \tag{5}$$

with $c_{\max}$ the field's own running maximum. Material that will not fit is *dropped and counted* in `apop_spill`, so the conservation law reports its own error rather than violating itself silently. Both restrictions are gated (G4, G5).

A death RATE, not a death set: with N live cells, $|\{\text{marked}\}| \leq \phi N$, and when the cap binds the worst candidates are taken by the mode's own severity. Geometric modes are exempt, because a set chosen once has no flux.

a_X, u_X	activator, substrate of species X	*per cell*, INTENSIVE. A concentration, so it is meaningful to compare between cells of different size
`chan`	species selector	which `chem` pair an operator owns. Default 0, so every pre-existing spec is unchanged
D_a^X, D_u^X	diffusivities	the RATIO sets the Turing wavelength; the pair sets the speed
ρ	growth baseline	tip:body is $(\rho+1):\rho$. $\rho = 0$ is an infinite ratio, which P1 refuses
x_{sw}, n	gate threshold, sharpness	a FRACTION of that field's own maximum, never absolute
s, c	`shrink_rate`, `critical_frac`	per tick, and a fraction OF v_{ref}
v_{ref}	reference volume	the SEED-TIME median. Never recomputed: recomputing it after deaths moves every threshold in the model
τ_{clear}	clearing time	INTENSIVE: a property of (s, c) alone, not of the tissue
ϕ	`max_mark_frac`	fraction of LIVE cells under sentence at once — bounds a FLUX
n_{apop}	deaths	EXTENSIVE: scales with the tissue, so never comparable between runs of different size without dividing by N
`apop_spill`	unplaced material	EXTENSIVE. The conservation law's own error term

3. The Plexus2 container

No new set. What is new is a *family* the vesicle lacked — *Die* — and a second species inside one state buffer. Three claims about the container have to hold, and each is a gate:

1. **Two instances of one operator are ADDITIVE, not overwriting.** `cell_react` returns a delta over the whole `chem` tensor and writes zeros outside the pair its `chan` names. Without that, a second instance overwrites the first and "two species" is a name for one. (G7)
2. **`apoptosis_3d` is `structural`, not `lateral`.** It changes how many entities exist, so it may not be a pure state map: it renumbers faces, and every per-face array and every *pending delta* must be permuted with it. (G3)
3. **A named population is a seed window, one level up.** `list`, `band` and `cone` choose a population ONCE. Re-evaluated every frame, a death sentence becomes a death rate — the same argument the paper makes for seeding, where an initial condition re-applied every step is a moving boundary condition. (G6)

operator	**kind / family**	**reads**	**mechanism**	**gate**
`seed_cell_rd`	seed / growth	—	seeds its own columns	G7
`cell_diffuse`	lateral / fields	`chem`, adjacency	per-column diffusivity vector	G7
`cell_react`	lateral / fields	`chem`	Eq. (1), zeros outside its pair	G7
`grow_3d`	structural / growth	`chem`	Eq. (3)	G8, G9
`apoptosis_3d`	**structural / die**	`chem`, mesh	Eqs. (4)–(5)	G1–G6
`reconnect_t1_3d`	rewire / topology	mesh	sheds neighbours — REQUIRED	G2

Schedule. Death runs *between growth and the mechanics*: growth sets the targets, death overrides them for the sentenced, and the relaxation sees both in one tick, so a cell extruded this frame has its hole closed this frame. Measured, same composition with death appended after the recorder instead: grip 0.031 against 0.110. There is no `substep_dt` block — this vesicle integrates at one rate and `shape_energy_3d` carries `relax_iters` internally.

4. The gates

Thresholds set before the run. The tiers are the paper's: **bookkeeping** asks whether the code does what the operator says, **closed form** whether it reproduces physics it was given, and **measurement** whether it agrees with something observed in cells.

	gate	threshold, before	measured
tier 1 — bookkeeping			
G1	every selector fires on a known population and NOT on a uniform field	all cases	**19/19**
G2	$\chi = 2$ at every frame with death running	exact	**2, all frames**
G3	a topology change permutes pending DELTAS, not only state	activator ≥ 0	**0** (was -1.04×10^{10})
G4	a bequest goes only to neighbours above the extrusion threshold	no divisor $< c\,v_{\mathrm{ref}}$	**P4, P12 cleared**
G5	a bequest leaves the recipient inside the integrator's basin	no NaN, activator ≥ 0	**RE-RUNNING** —
G6	a named population is chosen ONCE; only state modes re-evaluate	one death per named cell	**1, was 7**
G7	two species never write each other's columns	$\{0,1\}$ and $\{2,3\}$	**exact, non-uniforn**
G8	`inhib_chan` absent reproduces a pre-change run	metrics identical	**RUNNING** — §6 I
G9	growth $\to 0$ as the inhibitor saturates	$< 5\%$ of uninhibited	$0.978 \to 0.014$
G10	the cap admits at most ϕN, worst first, and drains	exact	**8/100; took worst;**
G11	n_{apop} equals the cells removed	exact, death-only composition	$400 - N_{\mathrm{final}}$
tier 2 — closed form			
G12	a collapse is a T2: $V-2$, $E-3$, $F-1$	exact, else refuse	**refuses $k > 3$, re-ch**
G13	clearing time equals $\ln c/\ln(1-s)$	within 1 tick	**NOT MEASUREI**
G14	the throughput lever is s, not ϕ	deaths $\propto 1/\tau_{\mathrm{clear}}$	$\phi\times50\to\times1.7$; $s\times3\to$
G15	a geometric chisel removes exactly its declared population	exact	**5/5 geometries**
G16	the second species' wavelength differs from the first	spot count differs	**NOT MEASUREI**
tier 3 — measurement			
G17	apoptotic rate matches an epithelium's	1–3% of cells per day	BLOCKED: no `unit`
G18	clearance time matches an extrusion in vivo	30–90 min	BLOCKED: no `unit`
G19	the two species' wavelengths bracket a real morphogen pair	—	BLOCKED: no `unit`

Eleven bookkeeping, five closed form, three measurement — and all three of the last are blocked for want of a `units:` block, the same defect the `ecm` study found when "0.82 grid cells" turned out to be 15 μm. So this note can establish that the operators do what they say and reproduce the arithmetic they were given. **It cannot yet say anything about cells.** Declaring units is the single change that would move three gates from blocked to failable.

5. What the runs establish

Twenty-four exploratory runs on `r001_03` and fifteen dedicated gate runs. Three results are properties of the operators rather than of any morphology, and are the ones this note rests on.

Position dominates rate. Death patterned in SPACE deforms the body at a thirtieth of the cell cost of death patterned by rate:

run	deaths	cells	protr_f	invag_f	red_vol
`sc_crown` (a polar cap)	**113**	16056	**1.267**	**0.343**	**0.814**
`sc_waist` (an equatorial band)	332	14486	1.174	0.328	0.807
`tsd_max` (the highest rate)	**3460**	2895	1.140	0.111	0.957
control (no death)	0	12692	1.095	0.120	0.959

Scattered death thins a ball into a smaller ball; a chosen population is a cut.

The throughput lever is τ_{clear}, not the cap. Fifty times ϕ moves the death count by 1.7×; tripling s at fixed ϕ moves it by 4.7×. The queue does not fill, so what binds is how fast a sentenced cell reaches the extrusion threshold (G14).

Inhibition is monotone and is a brake. 12692 → 6920 → 4859 → 4154 cells as `inhib_sw` tightens, with `protr_final` falling 1.095 → 1.020. Eq. (3) asserts a multiplicative shutdown and the tissue size follows it; the shape does not sharpen.

6. Open items

	item	what closing it requires	state
G5	the conservation bound holds	`tsd_cap10` and `tsd_cap25` complete with a finite activator and `apop_spill` reported.	running
G8	the inhibitor's null is exact	two replicates of one untouched spec. If they differ from each other, the engine is not deterministic and no per-run comparison in the campaign is exact; if they agree, the channel change perturbed a run that does not use it.	running
G13	$\tau_{\text{clear}} = \ln c / \ln(1-s)$	the death frame measures shrink time PLUS shedding time, and the formula is the first term only. Separating them needs the frame each marked cell crosses $c\,v_{\text{ref}}$ recorded beside the frame it is extruded.	**open**
G16	B's wavelength is coarser than A's	FAILS as stated: at $\chi = 0.4$ species B holds 15 spots against A's 2, so B is the FINE map. The requirement behind it — that the two maps differ — holds at 2.2× in spacing.	**closed, failed**
G16c	the pair keeps both wavelengths	per-species pattern metrics. Every chemistry metric read `chem[h0]`; the second species is now measured on `chan 2` and reported under `b_` keys.	running
I4	the slow ladder, unconfounded	three runs with `inhib_chan` absent, so death speed is the only variable.	running
I8	`apop_spill` is near zero on a healthy run	any completed death run: the term is recorded per frame and carried into the summary.	running

7. What this licenses, and what it does not

Licensed. Two RD species coexist in one buffer without leaking (G7), which is the structural claim the whole addition rests on. The Die family is sound as bookkeeping: selectors fire where they should and nowhere else (G1), the sheet stays closed (G2), a topology change carries its pending deltas (G3), the count is exact (G11), the rate is a rate (G10), and a named population is a seed window rather than a per-frame rule (G6). Growth can be switched off (G9), and the collapse is a genuine T2 (G12).

Not licensed. G5 is the one that matters most: until it closes, the operator can drive the chemistry out of its basin and a run's numbers are then about an integrator. **G13** means the clearing time — the quantity every death experiment is denominated in — is asserted and not measured. **No gate is a measurement** (G17–G19 blocked for want of a `units:` block), so nothing here has been compared with a cell.

The mechanism should not enter the loop until G5, G8 and G13 close. G5 because a mechanism that can NaN the chemistry will produce runs the campaign records as evidence; G8 because an operator that changes runs which do not use it corrupts every comparison in the ledger; G13 because the clearing time is the quantity every death experiment is denominated in. G16 and I4/I5 are about the experiments, not the operators, and can close afterwards.

8. References

References

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